

1. Prerequisites

- A running Kubernetes cluster (minikube, kind, EKS, GKE, AKS, etc.).
 - `kubectl` configured and working.
 - `helm` installed (we'll use Helm charts for easy deployment).
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2. Install Prometheus

Prometheus scrapes metrics (CPU, memory, custom app metrics).

```
helm repo add prometheus-community https://prometheus-  
community.github.io/helm-charts  
helm repo update
```

Install Prometheus (with node-exporter, kube-state-metrics, etc.):

```
helm install prometheus prometheus-community/kube-prometheus-stack -n  
monitoring --create-namespace
```

This installs **Prometheus + Grafana + Alertmanager** together.

👉 Grafana will already be included here.

3. Access Grafana

Check Grafana pod:

```
kubectl get pods -n monitoring
```

Get Grafana admin password:

```
kubectl get secret prometheus-grafana -n monitoring -o  
jsonpath="{.data.admin-password}" | base64 --decode ; echo
```

Port-forward Grafana:

```
kubectl port-forward svc/prometheus-grafana -n monitoring 3000:80
```

Open in browser: <http://localhost:3000>

Login:

- Username: admin
- Password: (the one you just decoded)

4. Install Loki

Now let's add Loki for logs.

```
helm repo add grafana https://grafana.github.io/helm-charts
helm repo update
```

Install Loki **and** Promtail (agent that pushes logs to Loki):

```
helm install loki grafana/loki-stack -n monitoring
```

This deploys:

- **Loki** → log storage
- **Promtail** → log collector (reads from Kubernetes pods and sends to Loki)

5. Configure Grafana Datasources

Now connect Grafana to Loki.

1. Go to Grafana UI → **Connections** → **Data sources** → **Add data source**.
2. Choose **Loki**.
3. URL → `http://loki:3100` (inside cluster),
or if using port-forward:
4. `kubectl port-forward svc/loki -n monitoring 3100:3100`

Then URL = `http://localhost:3100`.

5. Click **Save & Test**.

👉 You already have **Prometheus datasource** auto-installed from kube-prometheus-stack.

6. Dashboards

- Import pre-built dashboards:
 - Prometheus: Node Exporter / Kubernetes cluster monitoring.
 - Loki: Log panels.

Go to Grafana → Dashboards → Import → paste ID from Grafana.com dashboards.

Example IDs:

- **Kubernetes / Prometheus metrics:** 6417
 - **Loki / Promtail logs:** 13639
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7. Query Logs & Metrics

- In Grafana, create a new dashboard.
 - Add a panel:
 - **Prometheus datasource** → query `rate(container_cpu_usage_seconds_total[5m])`
 - **Loki datasource** → query `{job="varlogs"} |~ "error"`
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☒ At this point you'll have:

- Prometheus → metrics
- Loki → logs
- Grafana → dashboards with both logs + metrics